

Performance Testing

Project Testing · AI/ML and GenAI Track

Note: the "Measured" column is left blank on purpose. Run the tests on your own machine and fill in real values; do not submit numbers you have not measured.

Where time goes: most latency comes from the Gemini API calls (two on plan generation: the plan and the tip; one per feedback update). Local work such as validation, SQLite reads/writes and template rendering is small by comparison.

Performance test plan

Test	Method	Metric	Target	Measured
Home page load	Browser dev tools or curl on GET /	Response time	< 200 ms	
Health check	curl GET /health	Response time	< 100 ms	
Plan generation	Submit the form 10 times with different goals	Average and max time	< 15 s typical	
Feedback update	Submit feedback 10 times	Average and max time	< 15 s typical	
Concurrent users	Locust or k6 with 5–10 users on /	Error rate, p95 latency	0 errors on static pages	
AI failure handling	Use an invalid API key	Behaviour	Error page (HTTP 502), no crash	
Retry behaviour	Simulate 429/503 responses	Attempts made	Up to 3 with backoff	
Database scale	Insert 1,000 users, open /view-all-users	Page load time	Usable	

Functional test cases

ID	Test	Expected result
TC1	Valid form submission	Plan and tip shown; record saved
TC2	Age 12 or 101	Validation error page
TC3	User ID with spaces or symbols	Validation error page
TC4	Feedback for an unknown user ID	404 page: user not found
TC5	Feedback for a valid user	Revised plan shown and saved
TC6	Repeated feedback	Each revision builds on the previous one
TC7	Missing or invalid API key	Friendly AI-service error page
TC8	Re-submit the same user ID	Record reset with a new plan
TC9	Delete a user	User removed; redirect to admin list

Known issue found in code review

The routes `/view-all-users` and `/delete-user/{user_id}` are not protected by the `require_admin` dependency, although the README describes the dashboard as password-protected. Adding `dependencies=[Depends(require_admin)]` to these routes fixes it. Fix this before deployment and re-test.